Reinforcement Learning

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1 Introduction

Imagine a problem where we have an *agent* that interacts with an *environment* and *we* want it to achieve a goal without explicitly telling it what to do?

We could assign a *numerical return* to the outcomes with the largest value for the goal we would like the agent to achieve and simply tweak the agent to take *actions* (out of the set of possible actions at any given time) so as to maximize the return.

The return itself could come from rewards sprinkled along the way (think stock market returns) or from a final reward at the last step (checkmate!). In practice, given a goal, one has to decide on numerical rewards and compose a final return from those rewards that is consistent with the goal.

Let's make this concrete. Then a sequence of states and actions are written as

$$(S_0, A_0, \cancel{R_0}), (S_1, A_1, R_1), (S_2, A_2, R_2), \dots (S_T, \cancel{A_T}, R_T) \tag{1}$$

Here, we start in a state S_0 (which represents the *state of the environment as well as the internal state of the agent*) and may be only partially visible to the agent. The agent then takes the action A_0. This results in reward which we *choose to label* R_1. This explains why R_0 is cancelled out since the game has just started. This goes on till the game terminates at time T. Note it is not necessary that a game terminates.

This process is often made more complicated in practice because of inherent stochasticity in the transition of states after an action is taken as well as in the rewards generated. So in the notation above

- The game may start in a state $S_0 \sim p(S_0)$
- After the agent taken the action A_t the next state may be sample $S_{t+1} \sim p(S_{t+1}|S_t, A_t)$
- The reward is sampled $R_{t+1} \sim p(R_{t+1}|S_t, A_t)$

Note we have given in to the Markovian fantasy above in that we assume the next step depends only on the current state. In theory, this is not a big assumption, as one can always *define* the state to contain all the past information that is required. In practice though it is a strong assumption that is frequently made without much thought largely due to the complexity involved in the alternative assumption.

Now with this Markovian fantasy the thing the agent needs to do at time t is to decide what action to take in the state S_t. In order to do this the agent needs to have a *policy* from which actions are sampled $A_t \sim \pi(A_t|S_t)$. This may be a deterministic or stochastic policy. Now we come to the two main problems of Reinforcement Learning

- What are $p(R_{t+1}|S_t, A_t)$ and $p(S_{t+1}|S_t, A_t)$?

- What should $\pi(A_t|S_t)$ be?

The rest of the lecture will be discussing how to solve these problems but we briefly discuss the general ideas. First we need to get back to discussing the *return* - a numerical proxy for the goal instead of returns received immediately after an action is taken. The return would be some aggregating function of the rewards in the future of an action and for most problems the contribution of temporally distant reward ought to be smaller. For computational reason we take

$$G_t = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1} \tag{2}$$

Given a policy $\pi(A_t|S_t)$ we can *in theory* figure out $p(R_{t+1}|S_t, A_t)$ and $p(S_{t+1}|S_t, A_t)$ by simply sampling these repeatedly. Note these would be independent of the policy π as long as the policy leads to visiting the concerned state and take the corresponding action. We would also that way end up getting $p(G_t|S_t, A_t; \pi)$. This is the Monte Carlo method. For details see section 4.

There is also the bootstrapping method where we make initial guesses for $q_\pi(s, a)$ and take a few steps and update the initial guess by the reward obtained in the process and the best guess at the time. For details see section 6.

Once we have found the returns for a policy, we can then modify the policy to improve the returns. This is called policy improvement.

In practice one doesn't go all the way to complete the evaluation for a given policy before improving it. All the flavours of going back and forth between evaluation and improvement are termed *Generalized Policy Iteration*.

2 Markov Decision Processes

A MDP is a sequence of states, actions and rewards

$$(S_0, A_0, \cancel{R_0})), (S_1, A_1, R_1), (S_2, A_2, R_2) \dots (S_T, \cancel{A_T}, R_T) \tag{3}$$

that may or may not have a terminal state. If it does have a terminal state then we call the full run an *episode*.

The *process* part should be obvious but let's focus on the other terms. First let's discuss the term *Markov*. We make the Markovian assumptions that we can capture the process by just modelling $p(S_{t+1}|S_t, A_t)$ and $\pi(A_t|S_t)$. The former is the probability to transfer to a new state given a state and an action while the latter is the probability to take an action A_t given that the state is S_t. Strictly speaking, this is not an assumption as given a big enough *state space*, we can always capture the past states that matter in the current state. However, in practice this becomes an approximation because the efforts to enlarge the state space is based on feature engineering. Next, let's discuss the term *decision*. The term $\pi(A_t|S_t)$ is taken to be *tunable* and in that sense it is a decision being taken.

For an episodic MDP we can write the partition function

$$Z_\pi = \sum_{S_{0:T}, A_{0:T-1}} \left(\prod_{t=0}^{T-1} p(S_{t+1}|S_t, A_t)\pi(A_t|S_t) \right) p(S_0) \tag{4}$$

The continuing case is the formal $T \to \infty$ limit of the above.

This gives

$$\begin{aligned} \mathbb{E}_\pi[R_{\tau+1}] &= \sum_{S_{0:T}, A_{0:T-1}} R_{\tau+1} p(R_{\tau+1}|S_\tau, A_\tau) \left(\prod_{t=0}^{T-1} p(S_{t+1}|S_t, A_t)\pi(A_t|S_t) \right) p(S_0) \\ &= \sum_{S_{0:\tau}, A_{0:\tau}} R_{\tau+1} p(R_{\tau+1}|S_\tau, A_\tau) \prod_{t=0}^{\tau} p(S_{t+1}|S_t, A_t)\pi(A_t|S_t) p(S_0) \end{aligned} \tag{5}$$

The actions taken and states coming after the reward in question do not impact the expectation value. However, formally we will always take the product over all times as it doesn't make a difference.

So if the agent is about to decide which action to take at time t they should only care about $\mathbb{E}[R_{t+k}]\ \forall\ k\ \geq 1$. We can then say that the agent ought to decide based on

$$\sum_{k \geq 1} \mathbb{E}_\pi[R_{t+k}|S_t, A_t] \tag{6}$$

however if k is unbounded then this function is unbounded. We thus define a stochastic return

$$\begin{aligned} G_t &\coloneqq \sum_{k=1}^{\infty} \gamma^{k-1} R_{t+k} \\ &= R_{t+1} + \gamma G_{t+1} \end{aligned} \tag{7}$$

and the agent then takes actions based on

$$A_t = \underset{a}{\operatorname{argmax}}\ \mathbb{E}_\pi[G_t|S_t, a] \tag{8}$$

3 Value Functions

We define the *value function*

$$\begin{aligned} v_\pi(s) &\coloneqq \mathbb{E}_\pi[G_0|S_0 = s] \\ &= \sum_{t=0}^{\infty} \gamma^t \mathbb{E}_\pi[R_{t+1}|S_0 = s] \end{aligned} \tag{9}$$

and the *state-action value function*

$$
\begin{aligned}
q_\pi(s,a) &\coloneqq \mathbb{E}_\pi[G_0|S_0=s, A_0=a] \\
&= \sum_{t=0}^{\infty} \gamma^t \mathbb{E}_\pi[R_{t+1}|S_0=s, A_0=a]
\end{aligned}
\tag{10}
$$

Since the process is Markovian the transition probabilities are independent of the time step. Below we define

$$
p(s'|s,a) \coloneqq p(S_{t+1}=s'|S_t=s, A_t=a) \tag{11}
$$

and likewise other expressions.

We use

$$
\mathbb{E}[A] = \mathbb{E}[\mathbb{E}[A|B]] \tag{12}
$$

extensively.

We then get

$$
\begin{aligned}
v_\pi(s) &= \mathbb{E}_\pi[G_t|S_t=s] \\
&= \mathbb{E}_\pi[R_{t+1} + \gamma G_{t+1}|S_t=s] \\
&= \mathbb{E}_\pi[R_{t+1}|S_t=s] + \gamma \mathbb{E}_\pi[G_{t+1}|S_t=s] \\
&= \mathbb{E}_\pi[R_{t+1}|S_t=s] + \gamma \mathbb{E}_\pi[\mathbb{E}_\pi[G_{t+1}|S_{t+1}=s']|S_t=s] \\
&= \mathbb{E}_\pi[R_{t+1}|S_t=s] + \gamma \mathbb{E}_\pi[v_\pi(s')|S_t=s] \\
&= \sum_a \left[\mathbb{E}_\pi[R_{t+1}|S_t=s,a] + \gamma \mathbb{E}_\pi[v_\pi(s')|S_t=s,a]\right]\pi(a|s) \\
&= \sum_a \left[\sum_r r p(r|s,a) + \gamma \sum_{s'} v_\pi(s') p(s'|s,a)\right]\pi(a|s)
\end{aligned}
\tag{13}
$$

Similarly, we get

$$
\begin{aligned}
q_\pi(s,a) &= \mathbb{E}_\pi[G_t|S_t=s, A_t=a] \\
&= \mathbb{E}_\pi[R_{t+1}|S_t=s, A_t=a] + \gamma \mathbb{E}_\pi[G_{t+1}|S_t=s, A_t=a] \\
&= \mathbb{E}_\pi[R_{t+1}|S_t=s, A_t=a] + \gamma \mathbb{E}_\pi[\mathbb{E}_\pi[G_{t+1}|S_{t+1}=s', A_{t+1}=a']|S_t=s, A_t=a] \\
&= \mathbb{E}_\pi[R_{t+1}|S_t=s, A_t=a] + \gamma \mathbb{E}_\pi[q_\pi(s',a')|S_t=s, A_t=a] \\
&= \sum_r r p(r|s,a) + \gamma \sum_{s',a'} q_\pi(s',a') p(s',a'|s,a) \\
&= \sum_r r p(r|s,a) + \gamma \sum_{s',a'} q_\pi(s',a') \pi(a'|s') p(s'|s,a)
\end{aligned}
\tag{14}
$$

From these we see

$$\begin{aligned} v_\pi(s) &= \mathbb{E}_\pi[G_t|S_t = s] \\ &= \mathbb{E}_\pi[\mathbb{E}_\pi[G_t|S_t = s, A_t = a]] \\ &= \mathbb{E}_\pi[q_\pi(s,a)] \\ &= \sum_a q_\pi(s,a)\pi(a|s) \end{aligned} \tag{15}$$

Further we have

$$q_\pi(s,a) = \sum_r rp(r|s,a) + \gamma \sum_{s'} v_\pi(s')p(s'|s,a) \tag{16}$$

If these *were* to be known accurately, the agent can simply take action $A_t = \underset{a}{\operatorname{argmax}}\, q_\pi(S_t, a)$. However, easier said than done, there are two problems to be solved

- Actually find the correct values for $q_\pi(s,a)$
- Find the best $\pi_\star$ such that $\max_a q_{\pi_\star}(s,a) \geq \max_a q_\pi(s,a) \;\; \forall\, \pi, s$

We get for the optimal policy

$$v_\star(s) = \max_a \left[\sum_r rp(r|s,a) + \gamma \sum_{s'} v_\star(s')p(s'|s,a) \right] \tag{17}$$

$$q_\star(s,a) = \sum_r rp(r|s,a) + \gamma \sum_{s'} \left(\max_{a'} q_\star(s',a') \right) p(s'|s,a) \tag{18}$$

4 Monte Carlo

The expected reward is

$$\mathbb{E}_\pi[G_0|S_0] = \sum_{S_{1:T}, A_{0:T-1}} \left(\sum_{R_{1:T}} \sum_{\tau=0}^{T-1} \gamma^\tau R_{\tau+1} p(R_{\tau+1}|S_\tau, A_\tau) \right) \left(\prod_{t=0}^{T-1} p(S_{t+1}|S_t, A_t)\pi(A_t|S_t) \right) \tag{19}$$

If we repeatedly sample the MDP then the average value will give us the above on expectation. Furthermore, because it's a Markovian process, there is nothing special about the starting state so the multiple samplings of MDP will give us

$$q_\pi(s,a) = \mathbb{E}[G_0|S_0 = s, A_0 = a] \; \forall\; s, a \tag{20}$$

as long as the (s,a) pair are sampled sufficiently.

A very useful idea in Monte Carlo is that of *importance sampling*. Here, we may sample using policy b while learning the expectations for policy π.

Concretely, if we have two different policies π and b

$$\begin{aligned} E_\pi[R_{\tau+1}] &= \sum_{S_{0:T},A_{0:T-1}} R_{\tau+1} p(R_{\tau+1}|S_\tau, A_\tau) \left(\prod_{t=0}^{T-1} p(S_{t+1}|S_t, A_t)\pi(A_t|S_t) \right) p(S_0) \\ E_b[R_{\tau+1}] &= \sum_{S_{0:T},A_{0:T-1}} R_{\tau+1} p(R_{\tau+1}|S_\tau, A_\tau) \left(\prod_{t=0}^{T-1} p(S_{t+1}|S_t, A_t)b(A_t|S_t) \right) p(S_0) \end{aligned} \tag{21}$$

Then we get

$$E_\pi[R_{\tau+1}] = \mathbb{E}_b[\left(\prod_{t=0}^{\tau} \frac{\pi(A_t|S_t)}{b(A_t|S_t)} \right) R_{\tau+1}] \tag{22}$$

So we get

$$\begin{aligned} \mathbb{E}_\pi[G_0|S_0] &= \sum_{\tau=0}^{T-1} \gamma^\tau \mathbb{E}_b[\left(\prod_{t=0}^{\tau} \frac{\pi(A_t|S_t)}{b(A_t|S_t)} \right) R_{\tau+1}] \\ &= \sum_{S_{1:T},A_{0:T-1}} \left(\sum_{R_{1:T}} \sum_{\tau=0}^{T-1} \gamma^\tau R_{\tau+1} \left(\prod_{t=0}^{\tau} \frac{\pi(A_t|S_t)}{b(A_t|S_t)} \right) p(R_{\tau+1}|S_\tau, A_\tau) \right) \\ &\times \left(\prod_{t=0}^{T-1} p(S_{t+1}|S_t, A_t)\pi(A_t|S_t) \right) \end{aligned} \tag{23}$$

Note this is called *per-decision* importance sampling. Another version, one without a name, would have each R_t taken with $\prod_{n=0}^{T-1} \frac{\pi(A_n|S_n)}{b(A_n|S_n)}$. Since the actions after a reward do not matter, the extra terms just add to the variance.

5 Policy Gradient

5.1 From the Trajectory Return

We define

$$J_\theta = \sum_{S_{0:T},A_{0:T-1}} \left(\sum_{R_{1:T}} \sum_{\tau=0}^{T-1} \gamma^\tau R_{\tau+1} p(R_{\tau+1}|S_\tau, A_\tau) \right) \left(\prod_{t=0}^{T-1} p(S_{t+1}|S_t, A_t)\pi_\theta(A_t|S_t) \right) p(S_0) \tag{24}$$

We then have

$$
\begin{aligned}
\partial_\theta J_\theta &= \sum_{S_{0:T},A_{0:T-1}} \left(\sum_{R_{1:T}} \sum_{\tau=0}^{T-1} \gamma^\tau R_{\tau+1} p(R_{\tau+1}|S_\tau, A_\tau) \right) \\
&\times \left(\sum_{\eta=0}^{T-1} \partial_\theta \log \pi(A_\eta|S_\eta) \right) \left(\prod_{t=0}^{T-1} p(S_{t+1}|S_t, A_t)\pi_\theta(A_t|S_t) \right) p(S_0) \\
&= \sum_{S_{0:T},A_{0:T-1}} \left(\sum_{\eta=0}^{T-1} \sum_{R_{1:T}} \sum_{\tau=0}^{T-1} \gamma^\tau R_{\tau+1} (\partial_\theta \log \pi(A_\eta|S_\eta)) p(R_{\tau+1}|S_\tau, A_\tau) \right) \\
&\times \left(\prod_{t=0}^{T-1} p(S_{t+1}|S_t, A_t)\pi_\theta(A_t|S_t) \right) p(S_0) \\
&= \sum_{S_{0:T},A_{0:T-1}} \left(\sum_{\eta=0}^{T-1} \partial_\theta \log \pi(A_\eta|S_\eta) \gamma^\eta \sum_{R_{\eta+1:T}} \sum_{k=0}^{T-1-\eta} \gamma^k R_{\eta+k+1} p(R_{\eta+k+1}|S_{\eta+k}, A_{\eta+k}) \right) \\
&\times \left(\prod_{t=0}^{T-1} p(S_{t+1}|S_t, A_t)\pi_\theta(A_t|S_t) \right) p(S_0)
\end{aligned} \tag{25}
$$

We can use the relation between returns and rewards to rewrite the above as

$$
\partial_\theta J_\theta = \sum_{S_{0:T},A_{0:T-1}} \sum_{R_{1:T}} \left(\sum_{\eta=0}^{T-1} \gamma^\eta G_\eta \partial_\theta \log \pi(A_\eta|S_\eta) \right) \prod_{t=0}^{T-1} p(R_{t+1}|S_t, A_t) p(S_{t+1}|S_t, A_t)\pi_\theta(A_t|S_t) p(S_0) \tag{26}
$$

5.2 From the Value Function

Below we understand that $\pi(a|s)$ depends on θ parameterically.

We then have

$$
\begin{aligned}
\partial_\theta v_\pi(s) &= \partial_\theta \sum_a q_\pi(s,a)\pi(a|s) \\
&= \sum_a \partial_\theta \pi(a|s) q_\pi(s,a) + \sum_a \pi(a|s)\partial_\theta q_\pi(s,a) \\
&= \sum_a \partial_\theta \pi(a|s) q_\pi(s,a) + \gamma \sum_{a,s'} \pi(a|s) p(s'|s,a)\partial_\theta v_\pi(s') \\
&= \sum_a \partial_\theta \pi(a|s) q_\pi(s,a) + \gamma \sum_{s'} \partial_\theta v_\pi(s') p(s'|s)
\end{aligned} \tag{27}
$$

Thus, iterating the recursion and noting that a state x reached from s in k steps accumulates a factor γ^k, we define the (unnormalised) discounted visitation $p(x|s) := \sum_{k=0}^{\infty} \gamma^k p_k(x|s)$, where $p_k(x|s)$ is the k-step transition probability under π. This gives

$$\begin{aligned}\partial_\theta v_\pi(s) &= \sum_{x\in\mathcal{S}} p(x|s) \sum_a \partial_\theta \pi(a|x) q_\pi(x,a) \\ &= \mathbb{E}_\pi\left[\sum_a \partial_\theta \pi(a|x) q_\pi(x,a)\right] \\ &= \mathbb{E}_\pi\left[\sum_a \pi(a|x)\partial_\theta \left(\log \pi(a|x)\right) q_\pi(x,a)\right] \\ &= \mathbb{E}_\pi\left[\mathbb{E}_a[\partial_\theta \left(\log \pi(a|x)\right) q_\pi(x,a)]\right]\end{aligned} \tag{28}$$

6 Bootstrapping Methods

Bootstrap methods rely on *assuming* some initial value and updating it by partial traversals of the MDP. In general they are of the form

$$Q_{t+n}(S_t, A_t) = Q_{t+n-1}(S_t, A_t) + \alpha \text{ something} \tag{29}$$

where α is a hyperparameter controlling the EMA scale of updates and the extra part is the actual update. This is the part that will differ in the various methods described below.

The result would converge when

$$\mathbb{E}_b[\text{something}] = 0 \tag{30}$$

where b is the behaviour policy. The form of the correction would determine what the Q-values actually converge to.

6.1 On Policy aka SARSA

Let us define a generalized or partial return as

$$G_{t:t+n} \coloneqq \sum_{k=1}^{n} \gamma^{k-1} R_{t+k} + \gamma^n Q_{t+n-1}(S_{t+n}, A_{t+n}) \tag{31}$$

We see this is a natural generalisation of equation 7 where we take the actual returns for n steps and then the best estimate *at the time* of the state-action value of the resulting state after those n steps and the action taken thereafter.

Also note the recursive relation

$$G_{t:t+n} = R_{t+1} + \gamma G_{t+1:t+n} \tag{32}$$

and the boundary condition

$$G_{t+n:t+n} = Q_{t+n-1}(S_{t+n}, A_{t+n}) \tag{33}$$

The update rule is then

$$Q_{t+n}(S_t, A_t) \quad = \quad Q_{t+n-1}(S_t, A_t) + \alpha(G_{t:t+n} - Q_{t+n-1}(S_t, A_t)) \tag{34}$$

and the Q-values learnt are on-policy.

This $n = 1$ version of this is called SARSA and has the simple form

$$Q_{t+1}(S_t, A_t) \quad = \quad Q_t(S_t, A_t) + \alpha[R_{t+1} + \gamma Q_t(S_{t+1}, A_{t+1}) - Q_t(S_t, A_t)] \tag{35}$$

The general one can then be called the n step SARSA and writing more explicitly we get

$$Q_{t+n}(S_t, A_t) \quad = \quad Q_{t+n-1}(S_t, A_t) + \alpha[\sum_{k=1}^{n} \gamma^{k-1} R_{t+k} + \gamma^n Q_{t+n-1}(S_{t+n}, A_{t+n}) - Q_{t+n-1}(S_t, A_t)] \tag{36}$$

6.2 Off Policy aka Q-Learning

We us define

$$\rho_{t:t+n} = \prod_{\tau=t}^{min(t+n,T-1)} \frac{\pi(A_\tau|S_\tau)}{b(A_\tau|S_\tau)} \tag{37}$$

where π is the target policy and b is the behaviour policy.

We define the partial returns for off policy as

$$\tilde{G}_{t:t+n} = R_{t+1} + \gamma \rho_{t+1:t+1} \tilde{G}_{t+1:t+n} \tag{38}$$

with the boundary condition

$$\tilde{G}_{t+n:t+n} = Q_{t+n-1}(S_{t+n}, A_{t+n}) \tag{39}$$

The the n step SARSA can be replaced by

$$\begin{aligned} Q_{t+n}(S_t, A_t) &= Q_{t+n-1}(S_t, A_t) + \alpha[\tilde{G}_{t:t+n} - \rho_{t+1:t+n-1} Q_{t+n-1}(S_t, A_t)] \\ &= Q_{t+n-1}(S_t, A_t) \\ &+ \alpha \Bigg[\sum_{k=1}^{n} \gamma^{k-1} \rho_{t+1:t+k-1} R_{t+k} + \rho_{t+1:t+n} \gamma^n Q_{t+n-1}(S_{t+n}, A_{t+n}) \\ &- \rho_{t+1:t+n-1} Q_{t+n-1}(S_t, A_t) \Bigg] \end{aligned} \tag{40}$$

when we sample by policy b but want expectations as per π as the policy. Notice how we needing an importance sampling factor in front of $Q_{t+n-1}(S_t, A_t)$ in the TD error. This ensures that at convergence we have

$$\begin{aligned}
& \mathbb{E}_b[\sum_{k=1}^{n} \gamma^{k-1} \rho_{t+1:t+k-1} R_{t+k} + \rho_{t+1:t+n} \gamma^n Q_{t+n-1}(S_{t+n}, A_{t+n}) - \rho_{t+1:t+n-1} Q_{t+n-1}(S_t, A_t)] \\
= \ & \mathbb{E}_\pi[\sum_{k=1}^{n} \gamma^{k-1} R_{t+k} + \gamma^n Q_{t+n-1}(S_{t+n}, A_{t+n}) - Q_{t+n-1}(S_t, A_t)] \\
= \ & 0 \qquad (41)
\end{aligned}$$

which means that the Q-values being learnt are those of the target policy π.

Now notice that if we don't mind extra variance then we can instead write this as

$$Q_{t+n}(S_t, A_t) = Q_{t+n-1}(S_t, A_t) + \alpha \rho_{t+1:t+n} [\sum_{k=0}^{n-1} \gamma^k R_{t+1+k} + \gamma^n Q_{t+n-1}(S_{t+n}, A_{t+n}) - Q_{t+n-1}(S_t, A_t)] \qquad (42)$$

Now let's look at the $n = 1$ case. We get

$$Q_{t+1}(S_t, A_t) = Q_t(S_t, A_t) + \alpha [R_{t+1} + \gamma \frac{\pi(A_{t+1}|S_{t+1})}{b(A_{t+1}|S_{t+1})} Q_t(S_{t+1}, A_{t+1}) - Q_t(S_t, A_t)] \qquad (43)$$

If π is the greedy policy we get have $\pi(A_{t+1}|S_{t+1}) = \delta_{\underset{a}{\text{argmax}} \ Q_t(S_{t+1},a), A_{t+1}}$. This gives

$$Q_{t+1}(S_t, A_t) = Q_t(S_t, A_t) + \alpha [R_{t+1} + \gamma Q_t(S_{t+1}, A_{t+1}) \frac{\delta_{\underset{a}{\text{argmax}} \ Q_t(S_{t+1},a), A_{t+1}}}{b(A_{t+1}|S_{t+1})} - Q_t(S_t, A_t)] \qquad (44)$$

Furthermore, since

$$\begin{aligned}
& \mathbb{E}_b[Q_t(S_{t+1}, A_{t+1}) \delta_{\underset{a}{\text{argmax}} \ Q_t(S_{t+1},a), A_{t+1}} \frac{\pi(A_{t+1}|S_{t+1})}{b(A_{t+1}|S_{t+1})}] \\
= \ & \mathbb{E}_\pi[Q_t(S_{t+1}, A_{t+1}) \delta_{\underset{a}{\text{argmax}} \ Q_t(S_{t+1},a), A_{t+1}}] \\
= \ & \max_a Q_t(S_{t+1}, a) \qquad (45)
\end{aligned}$$

we can replace the stochastic term with it's average

$$Q_{t+1}(S_t, A_t) = Q_t(S_t, A_t) + \alpha [R_{t+1} + \gamma \max_a Q_t(S_{t+1}, a) - Q_t(S_t, A_t)] \qquad (46)$$

This has reduced variance than the sampled form of the expression.

6.3 Tree Backup

For this the recursive relation for the partial return is modified by the red and blue pieces below

$$\tilde{G}_{t:t+n} = R_{t+1} + \gamma \left(\sum_{a \neq A_{t+1}} Q_{t+n-1}(S_{t+1}, a)\pi(a|S_{t+1}) + \pi(A_{t+1}|S_{t+1})\tilde{G}_{t+1:t+n} \right) \tag{47}$$

Thus, while in Monte Carlo we put the full weight on the path taken by taking the red part as zero and the blue part as one, here we take the values of paths not taken into account as well.

Let's write down a couple of these

$$\tilde{G}_{t:t+1} = R_{t+1} + \gamma \sum_a \pi(a|S_{t+1}) Q_t(S_{t+1}, a) \tag{48}$$

and

$$\begin{aligned}\tilde{G}_{t:t+2} &= R_{t+1} + \gamma \sum_{a \neq A_{t+1}} \pi(a|S_{t+1}) Q_{t+1}(S_{t+1}, a) \\ &\quad + \gamma\pi(A_{t+1}|S_{t+1}) \left(R_{t+2} + \gamma \sum_a \pi(a|S_{t+2}) Q_{t+1}(S_{t+2}, a) \right)\end{aligned} \tag{49}$$

The update rule is

$$Q_{t+n}(S_t, A_t) = Q_{t+n-1}(S_t, A_t) + \alpha[\tilde{G}_{t:t+n} - Q_{t+n-1}(S_t, A_t)] \tag{50}$$

and the Q value learnt is as per the target policy.

A Some results about Importance Sampling

$$\begin{aligned}\mathbb{E}_b[f(A_t)\frac{\pi(A_t|S_t)}{b(A_t|S_t)}] &= \sum_a f(a)\frac{\pi(a|S_t)}{b(a|S_t)} b(a|S_t) \\ &= \sum_a f(a)\pi(a|S_t) \\ &= \mathbb{E}_\pi[f(A_t)]\end{aligned} \tag{51}$$

Therefore it is easy to see that

$$\mathbb{E}_b[c\frac{\pi(A_t|S_t)}{b(A_t|S_t)}] = c \tag{52}$$

where c is not a function of A_t.

B Bandits

B.1 Value Based Models

Suppose there is a process where one can choose among a set of actions $\{a\}$ that give rewards R then it makes sense to take actions that maximize the reward

$$q_*(a) = \mathbb{E}[R|A = a] \tag{53}$$

which is the expectation of the reward given the action chose is a.

We want to

- Find the true values for each action

$$Q_t(a) := \frac{\sum_{i=1}^{t-1} R_i \delta_{A_i,a}}{\sum_{i=1}^{t-1} \delta_{A_i,a}} \tag{54}$$

- Take the action with the max value

$$A_t := \underset{a}{\operatorname{argmax}}\, Q_t(a) \tag{55}$$

However, the tricky part is to do them together. For this we can

- Take an ϵ-greedy approach where we exploit with probability $1 - \epsilon$ and explore with probability ϵ,
- Use the upper-confidence-bound method (TODO: find the math behind this)

$$A_t := \underset{a}{\operatorname{argmax}} \left[Q_t(a) + c\sqrt{\frac{\ln(t)}{N_t(a)}} \right] \tag{56}$$

The computation of the running mean for rewards eqn 54 per action naively requires keeping track of all rewards but we can instead do

$$Q_{n+1} = Q_n + \frac{1}{n}[R_n - Q_n] \tag{57}$$

Furthermore if the problem is non-stationary we can instead have a fixed parameter α to exponential weight the prior reward

$$\begin{aligned} Q_{n+1} &= Q_n + \alpha[R_n - Q_n] \\ &= (1-\alpha)Q_n + \alpha R_n \\ &= \alpha R_n + (1-\alpha)\alpha R_{n-1} + (1-\alpha)^2 \alpha R_{n-2} + \dots \\ &= (1-\alpha)^n Q_1 + \alpha \sum_{i=1}^{n} (1-\alpha)^{n-i} R_i \end{aligned} \tag{58}$$

C Taking Averages For Monte Carlo Reinforcement Learning

C.1 Ordinary Importance Sampling

For Monte Carlo we can talk about two kinds of averages across episodes. The first is *ordinary importance sampling*

$$V(s) = \frac{\sum_{t \in \mathcal{T}(s)} \rho_{t:T(t)-1} G_t}{|\mathcal{T}(s)|} \tag{59}$$

The way to interpret this is that $\mathcal{T}(s)$ is the union of all time steps at which the state was s. If we are discussing first visit MC then this would be restricted to first visit. T(t) is the first time of termination following t. G_t are returns pertaining to time step t and $\rho_{t:T(t)-1}$ are the importance sampling ratios.

To show how this works we consider an example. We show how the values are updated for first visit and every visit MC for the first two episodes.

- Episode 1: $\{(s,a),(s',a',R_1),(s,a,R_2),(s_t,R_3\}$ We have

$$\begin{aligned} G_0 &= R_1 + \gamma R_2 + \gamma^2 R_3 & (60)\\ G_1 &= R_2 + \gamma R_3 & (61)\\ G_2 &= R_3 & (62)\end{aligned}$$

  - First Visit

$$\begin{aligned} q(s,a) &= G_0 \rho_{1,\infty} \\ &= (R_1 + \gamma R_2 + \gamma^2 R_3) \frac{\pi(a'|s')\pi(a|s)}{b(a'|s')b(a|s)} & (63)\\ q(s',a') &= G_1 \rho_{2,\infty} \\ &= (R_2 + \gamma R_3) \frac{\pi(a|s)}{b(a|s)} & (64)\end{aligned}$$

  - Every Visit

$$\begin{aligned} q(s,a) &= \frac{G_2 + G_0 \rho_{1,\infty}}{2} \\ &= \frac{(R_1 + \gamma R_2 + \gamma^2 R_3)\frac{\pi(a'|s')\pi(a|s)}{b(a'|s')b(a|s)} + R_3}{2} & (65)\\ q(s',a') &= G_1 \rho_{2,\infty} \\ &= (R_2 + \gamma R_3) \frac{\pi(a|s)}{b(a|s)} & (66)\end{aligned}$$

- Episode 2: $\{(s', a'), (s', a', R_4), (s, \bar{a}, R_5), (s_t, R_6)\}$

$$
\begin{aligned}
G_0 &= R_4 + \gamma R_5 + \gamma^2 R_6 && (67)\\
G_1 &= R_5 + \gamma R_6 && (68)\\
G_2 &= R_6 && (69)
\end{aligned}
$$

  - First Visit

$$
\begin{aligned}
q(s, a) &= (R_1 + \gamma R_2 + \gamma^2 R_3) \frac{\pi(a'|s')\pi(a|s)}{b(a'|s')b(a|s)} && (70)\\
q(s', a') &= \frac{(R_2 + \gamma R_3)\frac{\pi(a|s)}{b(a|s)} + (R_4 + \gamma R_5 + \gamma^2 R_6)\frac{\pi(a'|s')\pi(\bar{a}|s)}{b(a'|s')b(\bar{a}|s)}}{2} && (71)\\
q(s, \bar{a}) &= R_6 && (72)
\end{aligned}
$$

  - Every Visit

$$
\begin{aligned}
q(s, a) &= \frac{(R_1 + \gamma R_2 + \gamma^2 R_3)\frac{\pi(a'|s')\pi(a|s)}{b(a'|s')b(a|s)} + R_3}{2} && (73)\\
q(s', a') &= \frac{(R_2 + \gamma R_3)\frac{\pi(a|s)}{b(a|s)} + (R_4 + \gamma R_5 + \gamma^2 R_6)\frac{\pi(a'|s')\pi(\bar{a}|s)}{b(a'|s')b(\bar{a}|s)} + (R_5 + \gamma R_6)\frac{\pi(\bar{a}|s)}{b(\bar{a}|s)}}{3} && (74)\\
q(s, \bar{a}) &= R_6 && (75)
\end{aligned}
$$

C.2 Ordinary Importance Sampling: Per-importance importance sampling

At this point it makes sense to *re-write the above after dropping the importance sampling ratios after the reward* as they do not make an effect on expectation but increase variance.

- Episode 1: $\{(s, a), (s', a', R_1), (s, a, R_2), (s_t, R_3\}$

  - First Visit

$$
\begin{aligned}
q(s, a) &= R_1 + \gamma R_2 \frac{\pi(a'|s')}{b(a'|s')} + \gamma^2 R_3 \frac{\pi(a'|s')\pi(a|s)}{b(a'|s')b(a|s)} && (76)\\
q(s', a') &= R_2 + \gamma R_3 \frac{\pi(a|s)}{b(a|s)} && (77)
\end{aligned}
$$

  - Every Visit

$$
\begin{aligned}
q(s, a) &= \frac{R_1 + \gamma R_2 \frac{\pi(a'|s')}{b(a'|s')} + \gamma^2 R_3 \frac{\pi(a'|s')\pi(a|s)}{b(a'|s')b(a|s)} + R_3}{2} && (78)\\
q(s', a') &= R_2 + \gamma R_3 \frac{\pi(a|s)}{b(a|s)} && (79)
\end{aligned}
$$

- Episode 2: $\{(s', a'), (s', a', R_4), (s, \bar{a}, R_5), (s_t, R_6)\}$
  - First Visit

$$q(s,a) = R_1 + \gamma R_2 \frac{\pi(a'|s')}{b(a'|s')} + \gamma^2 R_3 \frac{\pi(a'|s')\pi(a|s)}{b(a'|s')b(a|s)} \tag{80}$$

$$q(s',a') = \frac{R_2 + \gamma R_3 \frac{\pi(a|s)}{b(a|s)} + R_4 + \gamma R_5 \frac{\pi(a'|s')}{b(a'|s')} + \gamma^2 R_6 \frac{\pi(a'|s')\pi(\bar{a}|s)}{b(a'|s')b(\bar{a}|s)}}{2} \tag{81}$$

$$q(s,\bar{a}) = R_6 \tag{82}$$

  - Every Visit

$$q(s,a) = \frac{R_1 + \gamma R_2 \frac{\pi(a'|s')}{b(a'|s')} + \gamma^2 R_3 \frac{\pi(a'|s')\pi(a|s)}{b(a'|s')b(a|s)} + R_3}{2} \tag{83}$$

$$q(s',a') = \frac{R_2 + \gamma R_3 \frac{\pi(a|s)}{b(a|s)} + R_4 + \gamma R_5 \frac{\pi(a'|s')}{b(a'|s')} + \gamma^2 R_6 \frac{\pi(a'|s')\pi(\bar{a}|s)}{b(a'|s')b(\bar{a}|s)} + R_5 + \gamma R_6 \frac{\pi(\bar{a}|s)}{b(\bar{a}|s)}}{3} \tag{84}$$

$$q(s,\bar{a}) = R_6 \tag{85}$$

C.3 Weighted Importance Sampling

$$V(s) = \frac{\sum_{t \in \mathcal{T}(s)} \rho_{t:T(t)-1} G_t}{\sum_{t \in \mathcal{T}(s)} \rho_{t:T(t)-1}} \tag{86}$$

The way to interpret this is that $\mathcal{T}(s)$ is the union of all time steps at which the state was s. If we are discussing first visit MC then this would be restricted to first visit. T(t) is the first time of termination following t. G_t are returns pertaining to time step t and $\rho_{t:T(t)-1}$ are the importance sampling ratios.

To show how this works we consider an example. We show how the values are updated for first visit and every visit MC for the first two episodes.

- Episode 1: $\{(s, a), (s', a', R_1), (s, a, R_2), (s_t, R_3\}$
  - First Visit

$$q(s,a) = (R_1 + \gamma R_2 + \gamma^2 R_3) \tag{87}$$

$$q(s',a') = (R_2 + \gamma R_3) \tag{88}$$

  - Every Visit

$$q(s,a) = \frac{(R_1 + \gamma R_2 + \gamma^2 R_3)\frac{\pi(a'|s')\pi(a|s)}{b(a'|s')b(a|s)} + R_3}{\frac{\pi(a'|s')\pi(a|s)}{b(a'|s')b(a|s)} + 1} \tag{89}$$

$$q(s',a') = (R_2 + \gamma R_3) \tag{90}$$

- Episode 2: $\{(s', a'), (s', a', R_4), (s, \bar{a}, R_5), (s_t, R_6)\}$
  - First Visit

$$
\begin{aligned}
q(s,a) &= (R_1 + \gamma R_2 + \gamma^2 R_3) && (91)\\
q(s',a') &= \frac{(R_2+\gamma R_3)\frac{\pi(a|s)}{b(a|s)} + (R_4+\gamma R_5+\gamma^2 R_6)\frac{\pi(a'|s')\pi(\bar{a}|s)}{b(a'|s')b(\bar{a}|s)}}{\frac{\pi(a|s)}{b(a|s)} + \frac{\pi(a'|s')\pi(\bar{a}|s)}{b(a'|s')b(\bar{a}|s)}} && (92)\\
q(s,\bar{a}) &= R_6 && (93)
\end{aligned}
$$

  - Every Visit

$$
\begin{aligned}
q(s,a) &= \frac{(R_1+\gamma R_2+\gamma^2 R_3)\frac{\pi(a'|s')\pi(a|s)}{b(a'|s')b(a|s)} + R_3}{\frac{\pi(a'|s')\pi(a|s)}{b(a'|s')b(a|s)} + 1} && (94)\\
q(s',a') &= \frac{(R_2+\gamma R_3)\frac{\pi(a|s)}{b(a|s)} + (R_4+\gamma R_5+\gamma^2 R_6)\frac{\pi(a'|s')\pi(\bar{a}|s)}{b(a'|s')b(\bar{a}|s)} + (R_5+\gamma R_6)\frac{\pi(\bar{a}|s)}{b(\bar{a}|s)}}{\frac{\pi(a|s)}{b(a|s)} + \frac{\pi(a'|s')\pi(\bar{a}|s)}{b(a'|s')b(\bar{a}|s)} + \frac{\pi(\bar{a}|s)}{b(\bar{a}|s)}} && (95)\\
q(s,\bar{a}) &= R_6 && (96)
\end{aligned}
$$

C.4 Algorithms

C.4.1 Every Visit Algorithms

Here is the algorithm for on-policy MC prediction.

On-policy prediction

Initialize $Q(s,a) = \text{random}$ and $C(s,a) = 0 \ \forall\ s, a$
Loop over episodes:
  Using policy π generate the sequence $S_0, A_0, R_1, S_1, A_1 \ldots S_{T-1}, A_{T-1}, R_T, S_T$
  $G = 0$
  For t in $(T-1, T-2, \ldots 0)$:
    $G = \gamma G + R_{t+1}$
    $Q(S_t, A_t) = \frac{C(S_t,A_t)Q(S_t,A_t)+G}{C(S_t,A_t)+1}$
    $C(S_t, A_t) += 1$

The algorithm for off-policy ordinary importance sampling MC prediction is

Off-policy ordinary importance sampling prediction

Initialize $Q(s,a) = \text{random}$ and $C(s,a) = 0 \;\forall\; s,a$
Loop over episodes:
  Using policy b generate the sequence $S_0, A_0, R_1, S_1, A_1 \ldots S_{T-1}, A_{T-1}, R_T, S_T$
  $G = 0$
  $W = 1$
  For t in $(T-1, T-2, \ldots 0)$:
    $G = \gamma G + R_{t+1}$
    $Q(S_t, A_t) = \frac{C(S_t,A_t)Q(S_t,A_t)+WG}{C(S_t,A_t)+1}$
    $C(S_t, A_t) += 1$
    $W* = \frac{\pi(A_t|S_t)}{b(A_t|S_t)}$

The algorithm for off-policy weighted importance sampling MC prediction is

Off-policy weighted importance sampling prediction

Initialize $Q(s,a) = \text{random}$ and $C(s,a) = 0 \;\forall\; s,a$
Loop over episodes:
  Using policy b generate the sequence $S_0, A_0, R_1, S_1, A_1 \ldots S_{T-1}, A_{T-1}, R_T, S_T$
  $G = 0$
  $W = 1$
  For t in $(T-1, T-2, \ldots 0)$ while $W \neq 0$:
    $G = \gamma G + R_{t+1}$
    $Q(S_t, A_t) = \frac{C(S_t,A_t)Q(S_t,A_t)+WG}{C(S_t,A_t)+W}$
    $C(S_t, A_t) += W$
    $W* = \frac{\pi(A_t|S_t)}{b(A_t|S_t)}$

The algorithm for off-policy per decision ordinary importance sampling MC prediction is

Off-policy per decision ordinary importance sampling prediction

Initialize $Q(s,a) = $ random and $C(s,a) = 0 \;\forall\; s,a$
Loop over episodes:
 Using policy b generate the sequence $S_0, A_0, R_1, S_1, A_1 \ldots S_{T-1}, A_{T-1}, R_T, S_T$
 $G = 0$
 For t in $(T-1, T-2, \ldots 0)$:
  $G = \gamma G + R_{t+1}$
  $Q(S_t, A_t) = \frac{C(S_t,A_t)Q(S_t,A_t)+G}{C(S_t,A_t)+1}$
  $C(S_t, A_t) += 1$
  $G* = \frac{\pi(A_t\|S_t)}{b(A_t\|S_t)}$

C.5 Example

Suppose we have a state s and one terminal state t. There are only two actions left L and right R. The left takes the agent to the terminal state with probability q and return 1. It takes the agent to the state s with probability $1-q$. Suppose further the target policy has $\pi(L|s) = p$ and the exploration policy has $b(L|s) = \tilde{p}$. We evaluate the value and variance of the state under the first visit method. Then we have

$$\begin{aligned} \mathbb{E}_\pi[G_0] &= pq\sum_{k=0}^{\infty}(p(1-q))^k \\ &= \frac{pq}{1-p(1-q)} \end{aligned} \tag{97}$$

We also have

$$\begin{aligned} \mathbb{E}_\pi[G_0^2] &= pq\sum_{k=0}^{\infty}(p(1-q))^k \\ &= \frac{pq}{1-p(1-q)} \end{aligned} \tag{98}$$

If we compute this with importance sampling we get

$$\begin{aligned} \mathbb{E}_b[G_0\prod_{t=0}^{\infty}\frac{\pi(A_t|S_t)}{b(A_t|S_t)}] &= \tilde{p}q\sum_{k=0}^{\infty}(\tilde{p}(1-q))^k\left(\frac{p}{\tilde{p}}\right)^{k+1} \\ &= \frac{pq}{1-p(1-q)} \end{aligned} \tag{99}$$

which is the same as under the target policy.

We also get

$$\begin{aligned}\mathbb{E}_b[\left(G_0 \prod_{t=0}^{\infty} \frac{\pi(A_t|S_t)}{b(A_t|S_t)}\right)^2] &= \tilde{p}q \sum_{k=0}^{\infty} (\tilde{p}(1-q))^k \left(\frac{p}{\tilde{p}}\right)^{2(k+1)} \\ &= \frac{qp^2}{\tilde{p}} \sum_{k=0}^{\infty} \left(\frac{(1-q)p^2}{\tilde{p}}\right)^k \end{aligned} \tag{100}$$

This variance is divergent if $(1-q)p^2 \geq \tilde{p}$. In Barto and Sutton, they take $q = 0.1, p = 1$ and $\tilde{p} = 0.5$.

D Importance Sampling Example

Note that

$$v_\pi(s) = \mathbb{E}_b[\rho_{t:T-1} G_t | S_t = s] \tag{101}$$

where $\rho_{t:T-1} = \prod_{k=t}^{T-1} \frac{\pi(A_k|S_k)}{b(A_k|S_k)}$ and

$$q_\pi(s,a) = \mathbb{E}_b[\rho_{t+1:T-1} G_t | S_t = s, A_t = a] \tag{102}$$

Suppose we have a sequence $S_0, A_0, R_1, S_1, A_1, R_2, S_2, A_2, R_3, S_3$ where $S_0 = S_2 = s$ and $A_0 = A_2 = a$ and $S_1 = s', A_1 = a'$ then for ordinary importance sampling starting with $Q(S,A) = 0$ we get

$$\begin{aligned} q(s,a) &= \frac{G_2 + G_0 \rho_{1:2}}{2} \\ &= \frac{R_3 + (R_1 + \gamma R_2 + \gamma^2 R_3) \frac{\pi(A_1|S_1)\pi(A_2|S_2)}{b(A_1|S_1)b(A_2|S_2)}}{2} \end{aligned} \tag{103}$$

$$\begin{aligned} q(s',a') &= G_1 \rho_{2:2} \\ &= (R_2 + \gamma R_3) \frac{\pi(A_2|S_2)}{b(A_2|S_2)} \end{aligned} \tag{104}$$

whereas for weighted importance sampling we have

$$\begin{aligned} q(s,a) &= \frac{G_2 + G_0 \rho_{1:2}}{1 + \rho_{1:2}} \\ &= \frac{R_3 + (R_1 + \gamma R_2 + \gamma^2 R_3) \frac{\pi(A_1|S_1)\pi(A_2|S_2)}{b(A_1|S_1)b(A_2|S_2)}}{1 + \frac{\pi(A_1|S_1)\pi(A_2|S_2)}{b(A_1|S_1)b(A_2|S_2)}} \end{aligned} \tag{105}$$

$$\begin{aligned} q(s',a') &= G_1 \\ &= (R_2 + \gamma R_3) \end{aligned} \tag{106}$$

Now note that because of causality we have

$$\mathbb{E}_b[\rho_{t:T-1}R_{t+k}] = \mathbb{E}_b[\rho_{t:t+k-1}R_{t+k}] \tag{107}$$

so we could also simply take

$$q(s,a) = \frac{R_3 + (R_1 + \gamma R_2 \frac{\pi(A_1|S_1)}{b(A_1|S_1)} + \gamma^2 R_3 \frac{\pi(A_1|S_1)\pi(A_2|S_2)}{b(A_1|S_1)b(A_2|S_2)})}{2} \tag{108}$$

$$q(s',a') = (R_2 + \gamma R_3 \frac{\pi(A_2|S_2)}{b(A_2|S_2)}) \tag{109}$$